



**Design and Implementation of a Heart Rate Monitor Using Active Bandpass Filtering
(Official Documentation)**

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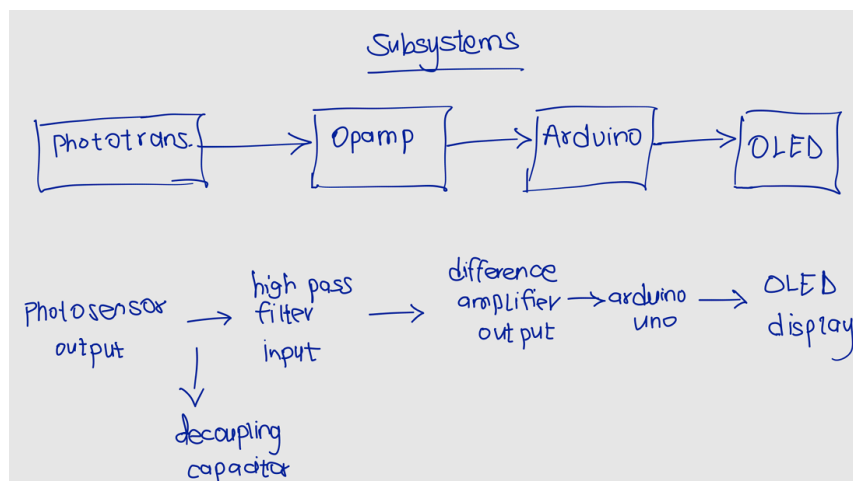
OVERVIEW

Heart rate is a vital metric used to assess physiological health, typically measured in Beats Per Minute (BPM). While electrocardiography (ECG) measures the electrical activity of the heart, this project focuses on a technique known as photoplethysmography (PPG), which is a non-invasive optical method that detects volumetric changes in blood circulation. The fundamental principle relies on the fact that blood absorbs infrared light; during systole (heart contraction), blood volume in the finger increases, absorbing more light, while during diastole (relaxation), volume decreases, allowing more light to reflect.

Objective

The objective of this project was to design and construct a non-invasive optical heart rate monitor based on the principle of Photoplethysmography (PPG). The system detects blood volume changes in the finger, conditions the analog signal to remove noise, and utilizes a microcontroller to calculate and display the Beats Per Minute (BPM) for a lay user.

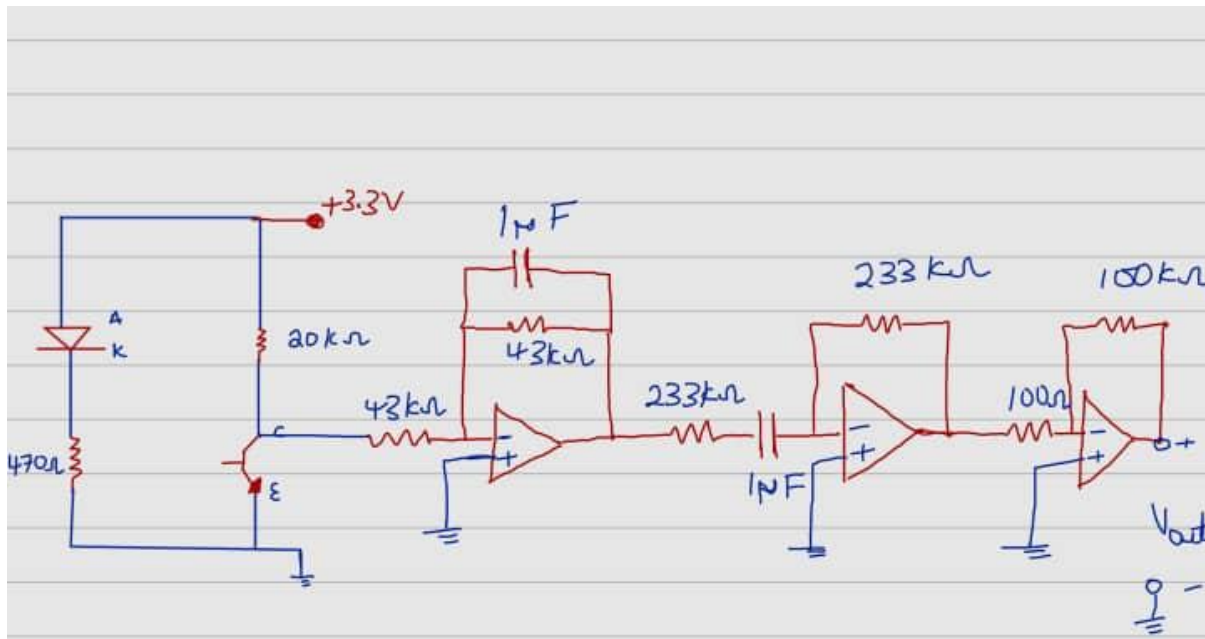
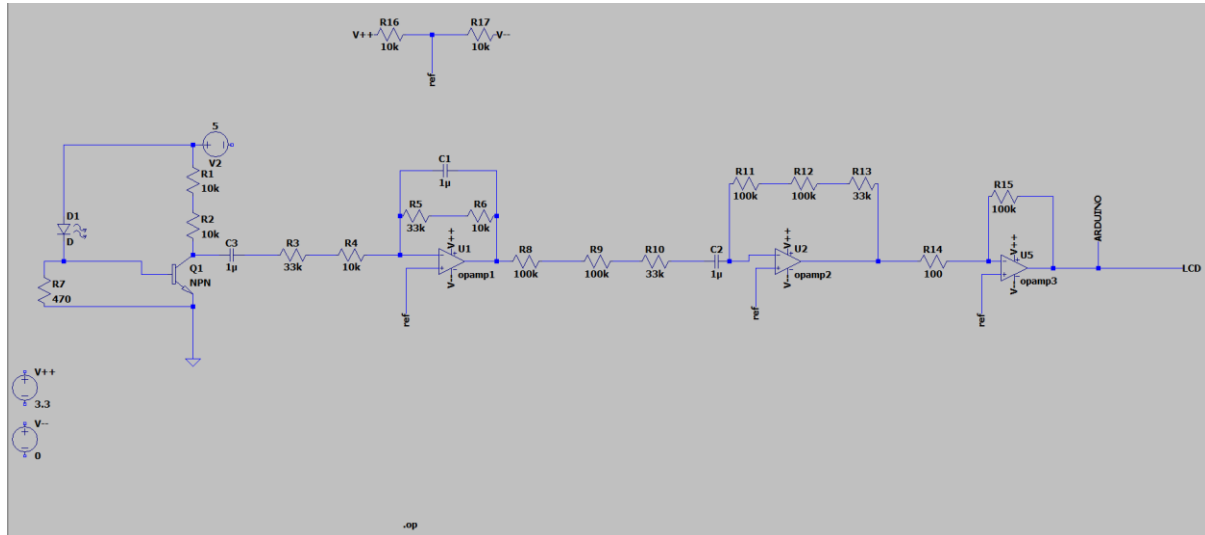
The system architecture follows a linear signal path. The signal originates at the phototransistor, which detects variations in light absorption caused by blood flow. This raw signal is passed through a high-pass filter to remove the DC offset, then into an amplification stage, followed by a low-pass filter to remove high-frequency noise. Finally, the conditioned analog signal is fed into the Arduino Uno, which processes the data and outputs the information to an LCD screen.



The raw signal from the PPG sensor contains a significant DC component caused by skin and tissue reflection, which is much larger than the AC signal of the actual heartbeat. To isolate the heartbeat, we utilized a bandpass filter configuration. We chose a bandpass filter rather than a simple low-pass or high-pass filter because the human heart rate falls within a specific frequency window.

We specifically implemented an Active Bandpass Filter rather than a passive one. Passive filters attenuate the signal, whereas an active filter, utilizing an op-amp, allows us to provide gain greater than one. This is crucial for "upscaling" the millivolt-level signal from the finger pulse into a voltage range (0-5V) that the Arduino can read accurately.

The amplification stage is integrated into this filter design. The gain of the amplifier is determined by the ratio of the feedback resistor R_{feedback} to the input resistor R_{input} . By using an input resistor of approximately 100Ω and a feedback resistor of $100k\Omega$, we achieve a gain of 100. This gain of 100 is sufficient to boost the weak finger pulse signal to a readable level for the microcontroller.



METHODOLOGY AND DESIGN CALCULATIONS

Frequency Domain Analysis

Our research established that the lowest resting heart rate is approximately 40 BPM (0.7 Hz) and the maximum is roughly 220 BPM (3.67 Hz). Therefore, the filter was designed to pass frequencies between 0.7 Hz and 3.5 Hz while rejecting everything else.

To design the filters, we first researched the frequency range of the human heart and converted it to Hz via a general formula $f = \frac{BPM}{60}$

Lower Limit (Diastole): 40 BPM.

Upper Limit (Systole): 220 BPM.

$$f_{\text{high}} = \frac{220}{60} = 3.67 \text{ Hz}$$

$$f_{\text{low}} = \frac{40}{60} = 0.67 \text{ Hz}.$$

Thus, the circuit was designed as a Bandpass Filter operating between 0.7 Hz and 3.67 Hz.

Target Heart Rate Band

$$f_{\text{low}} = \frac{40}{60} = 0.7 \text{ Hz} \quad f_{\text{high}} = \frac{220}{60} = 3.67 \text{ Hz}$$

This corresponds to the expected minimum and maximum human heart rates.

Low-Pass Filter

The low-pass filter is used to attenuate frequencies above 3.67 Hz.

$$f_{c1} = 3.67 \text{ Hz}$$

Using:

$$\omega_c = 2\pi f_c = \frac{1}{R_L C_L}$$

$$R_L = \frac{1}{2\pi(3.67)(1 \times 10^{-6})} \approx 43 \text{ k}\Omega$$

High-Pass Filter

The high-pass filter is used to attenuate frequencies below 0.7 Hz.

$$f_{c2} = 0.7 \text{ Hz}$$

$$R_H = \frac{1}{2\pi(0.7)(1 \times 10^{-6})} \approx 227 \text{ k}\Omega$$

Difference Amplifier Gain

$$\text{Gain} = \frac{R_f}{R_{in}} = \frac{100 \text{ k}\Omega}{100 \Omega} = 1000$$

DIGITAL LOGIC

The digital processing is handled by an Arduino Uno, but a significant portion of our project's technical complexity lies in the integration of the 16x2 LCD module. Unlike simple LED indicators, the LCD requires precise parallel communication. We configured the pinout manually, connecting the Register Select (RS) to pin 12 and the Enable (E) pin to pin 11, while the data lines D4 through D7 were mapped to digital pins 5, 4, 3, and 2 respectively.

A critical component in this setup is the potentiometer connected to the LCD's contrast pin (V0). This potentiometer acts as a voltage divider; without it, the text on the screen would be invisible because the contrast voltage must be finely tuned to make the pixels distinct from the background

The software logic is designed to sample, detect, and interpret the analog signal in real-time. The program begins by initializing the LCD and displaying a "Place finger" prompt. The core

measurement function operates on a strict 10-second timer. Inside this loop, the Arduino reads the voltage from Analog Pin A0.

Once the 10-second sampling period concludes, the system calculates the heart rate. The total number of beats counted during the interval is multiplied by 6 to extrapolate the Beats Per Minute (BPM). This calculated BPM is then passed to a custom function, `getBPMState`, which categorizes the user's heart rate into specific health ranges derived from our research.

The LCD displays the final BPM alongside a status message: "Resting" for values under 55, "Normal" for up to 70, "Light Act." for up to 100, "Moderate" for up to 140, "Intense" for up to 170, and "Danger" for higher values. This provides immediate, layman-friendly feedback on the user's physiological state.

RESULTS AND DISCUSSION

The final system successfully detects a finger pulse. The LCD provides user instructions ("Place Finger", "Hold Still"), displays a countdown timer, and outputs a stable BPM reading with the corresponding health status.

Implementation Challenges

During the construction phase, we encountered specific challenges:

The primary design challenge involved isolating the minute AC heartbeat signal (approximately 2mV–5mV) from the large DC component caused by skin and tissue reflection, as well as filtering out high-frequency noise.

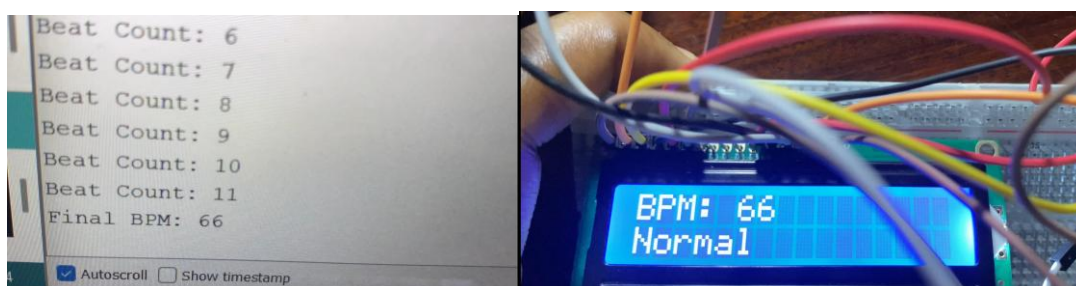
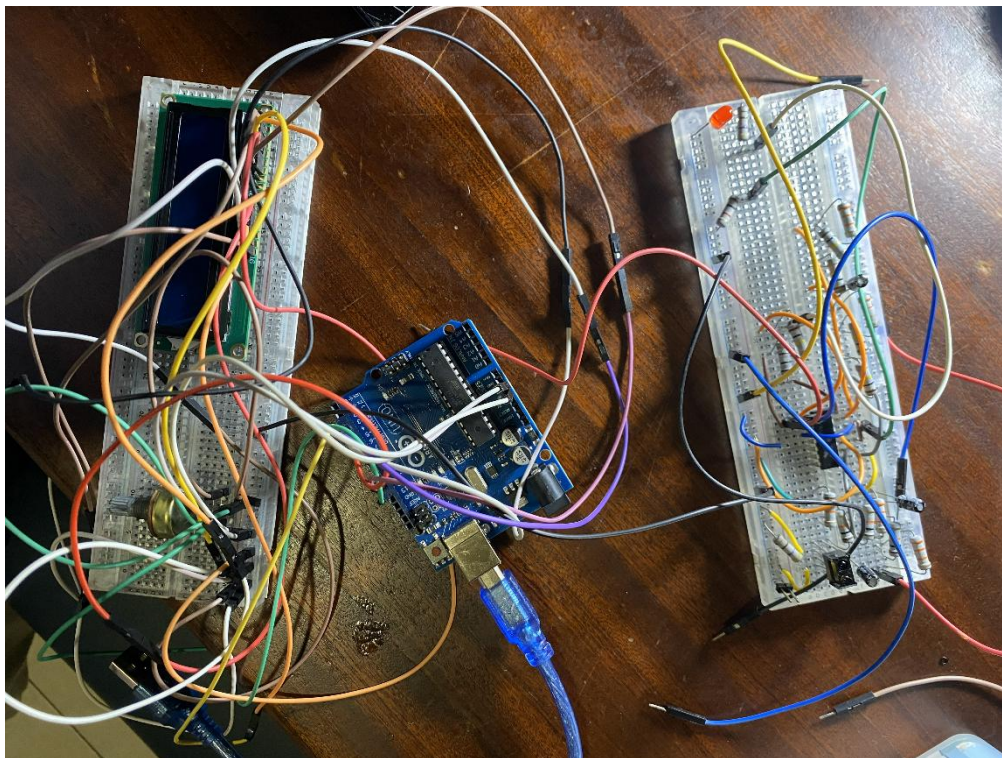
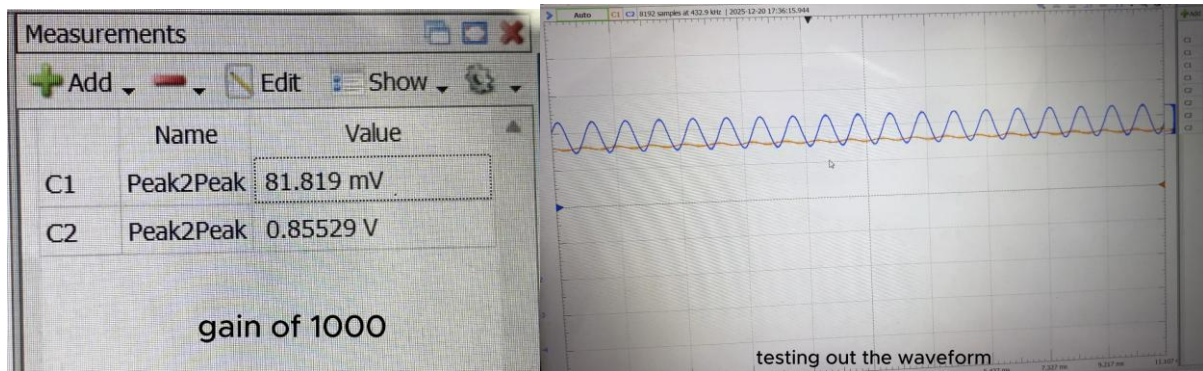
Initially, the Op-Amp output was insufficient to trigger the Arduino Digital Pin 6. This resulted in zero readings. We resolved this by switching to Analog Pin A0 and implementing a software threshold, allowing us to detect the 1.9V peaks successfully.

We also initially used a 100nF capacitor for the High Pass filter, which resulted in a cutoff frequency that was too high. We swapped this for a 1μF capacitor to correctly set the cutoff to 0.67 Hz.

CONCLUSION

Our project successfully showed the principles of instrumentation by combining analog signal conditioning (active filtering) with digital processing to build a functional device capable of monitoring heart rate in real-time. The final design met all specifications, including the frequency bandwidth of 0.7Hz–4Hz and the user-friendly display of health data.

APPENDIX



REFERENCES

- Lapitan, D. G., Rogatkin, D. A., Molchanova, E. A., & Tarasov, A. P. (2024). Estimation of phase distortions of the photoplethysmographic signal in digital IIR filtering. *Scientific Reports*, 14(1), 6546. <https://doi.org/10.1038/s41598-024-57297-3>